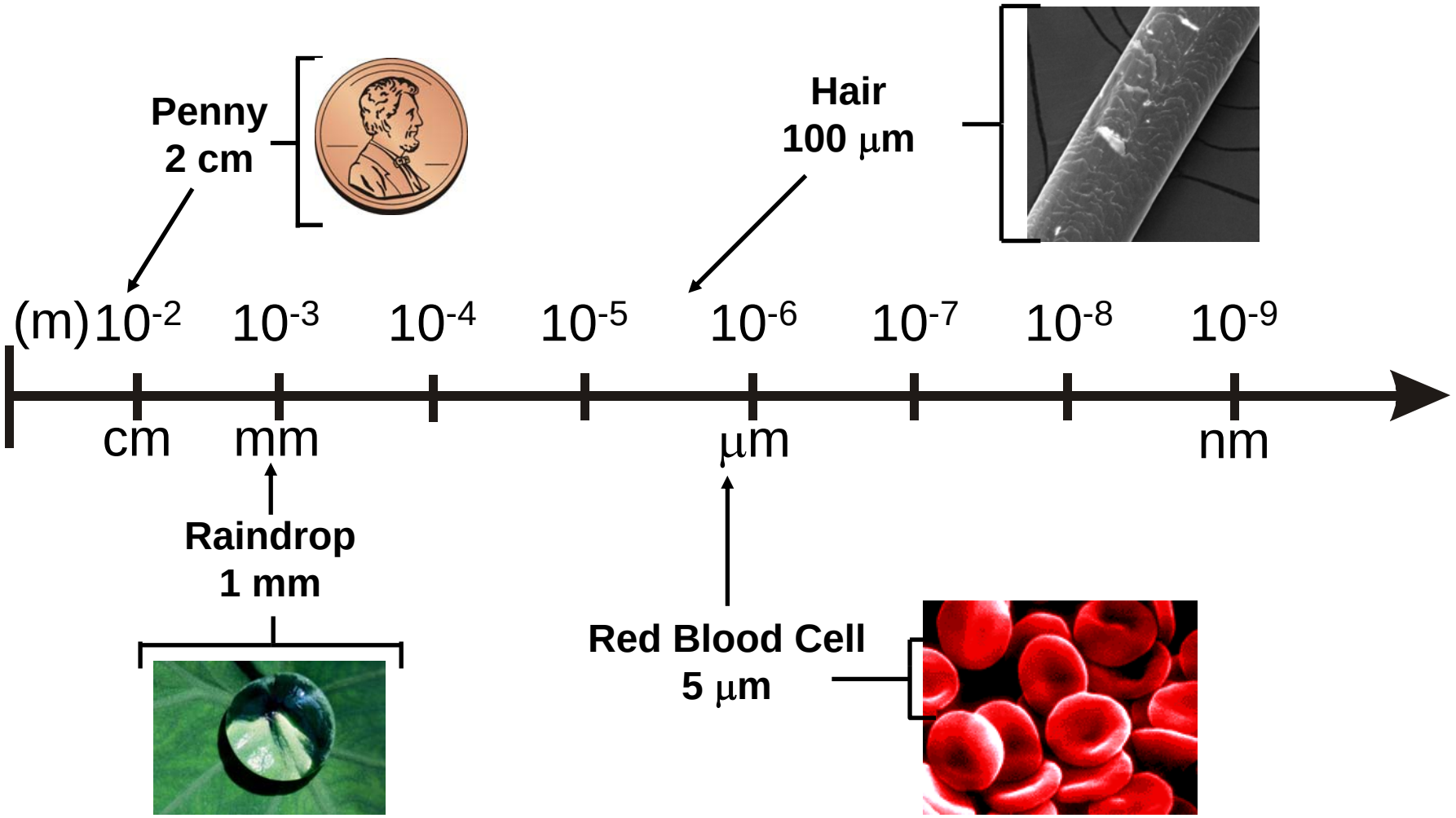
화학특강 (2009년 2학기)
소개

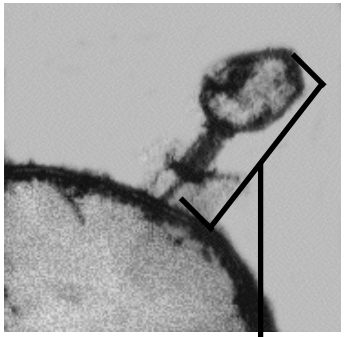
황현석

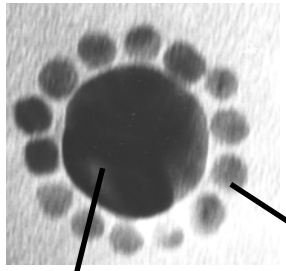
What Is A Nanometer?



Now Entering The NANO-ZONE

Bacteriophage
60-70 nm

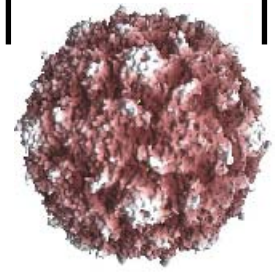




Gold Particles
13 nm & 50 nm

100 80 60 40 20 1

(nm)

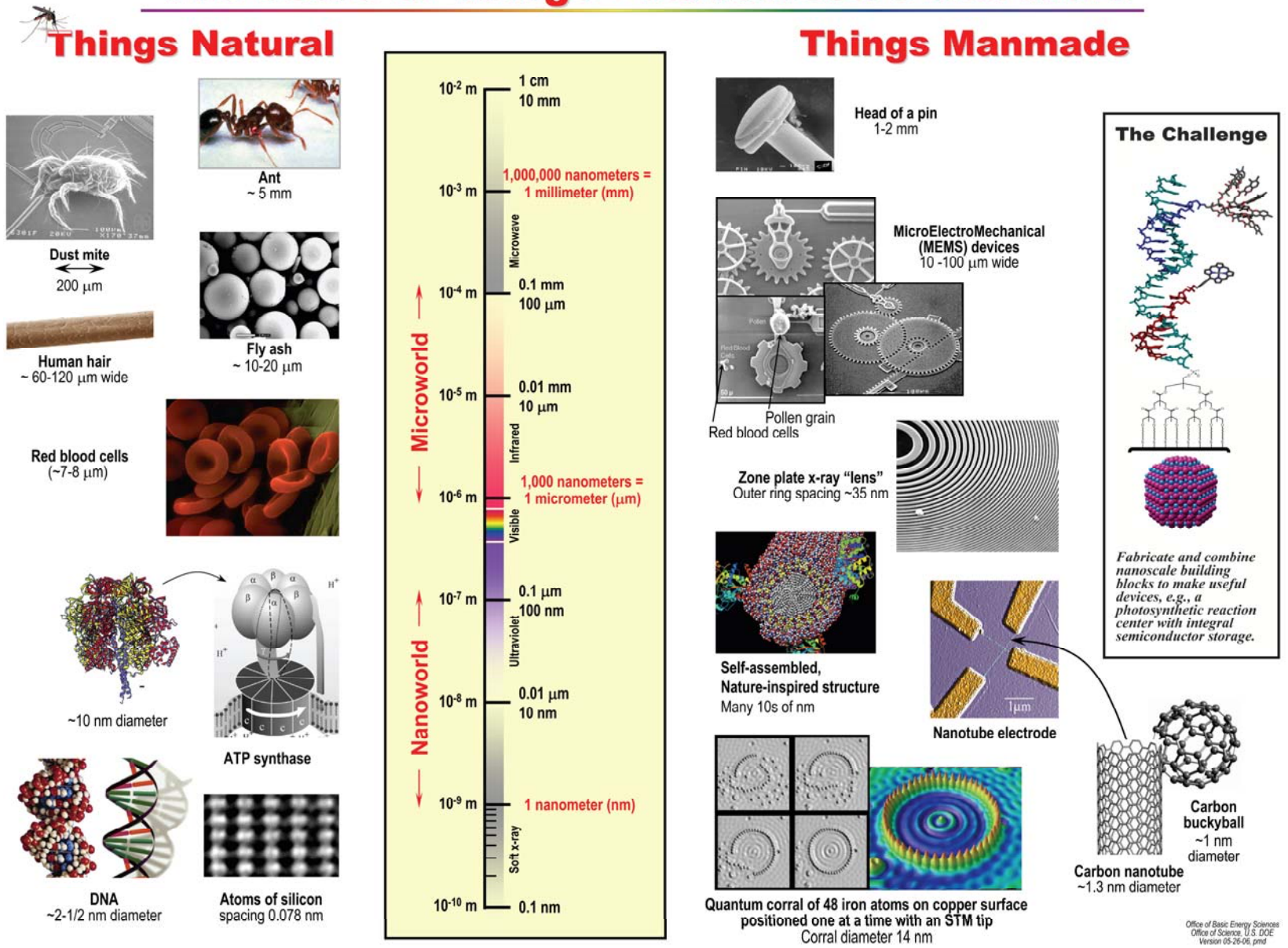


Flu Virus
100 nm

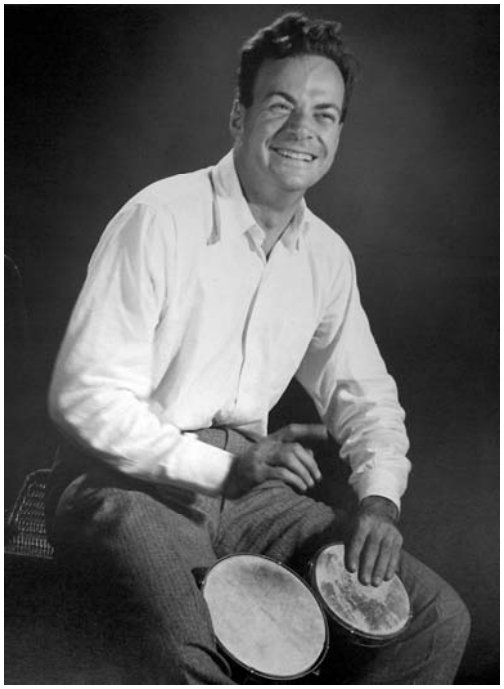


DNA
Diameter = 2 nm

What is Nanotechnology?



Origin of the term “Nanotechnology”



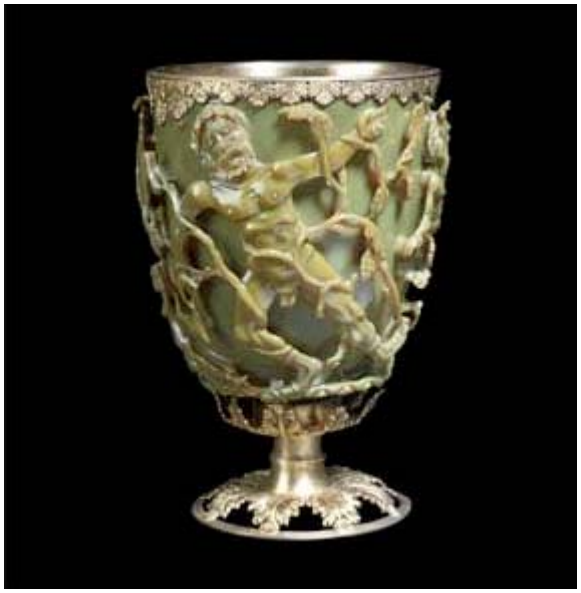
“There’s Plenty of Room at the Bottom”, the title of a talk given by physicist Richard Feynman



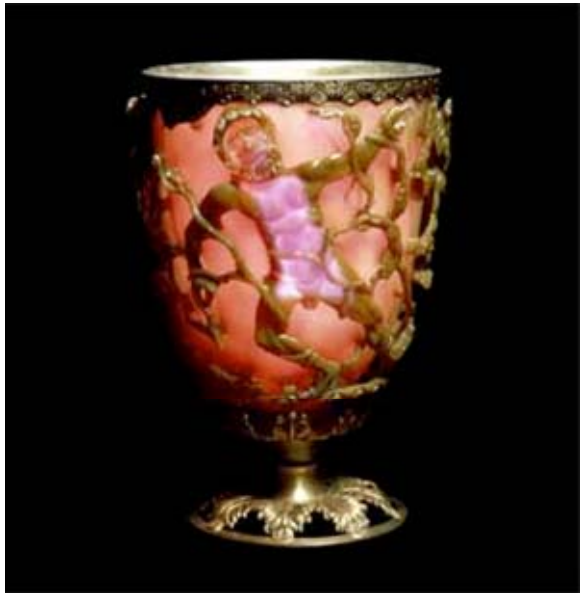
“**Nanotechnology** mainly consists of the processing of separation, consolidation, and deformation of materials by one atom or by one molecule.”
Norio Taniguchi, “On the Basic Concept of Nanotechnology”, *Proc. Intl. Conf. Prod.* London, Part II, 1974.

Nanotechnology Is Not a New Phenomenon

The Lycurgus Cup: 4th Century A.D.



Green = Reflected Light



Red = Transmitted Light

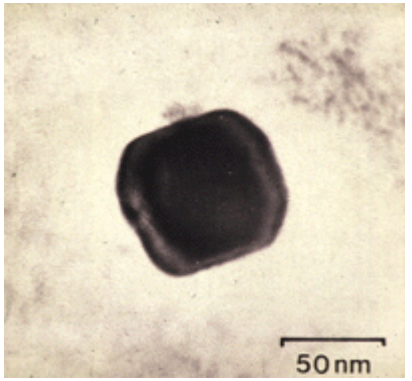
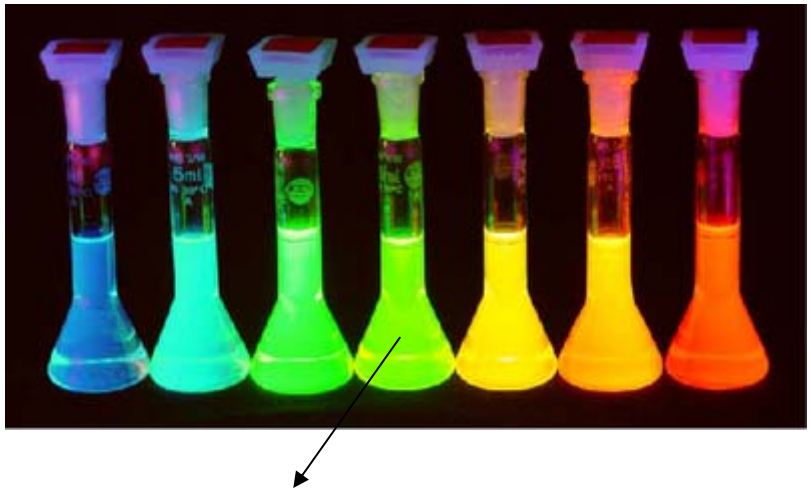
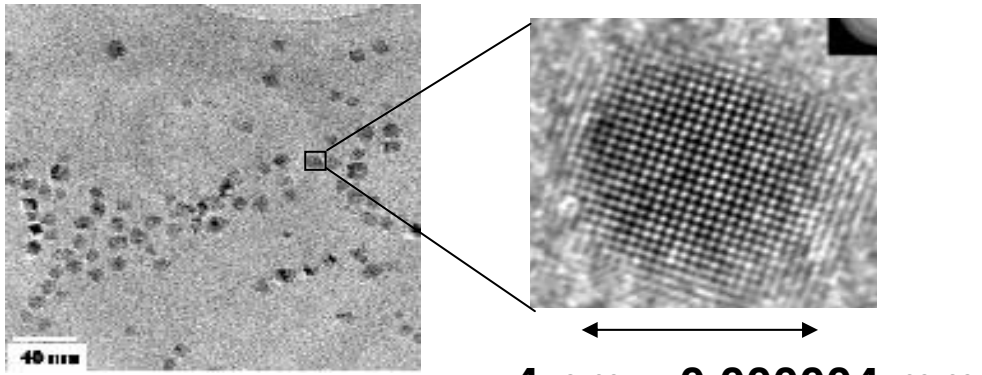


Image of silver/gold nanoparticle in the Lycurgus cup (300 ppmsilver, 40 ppm gold)

Quantum Dots and Their Applications

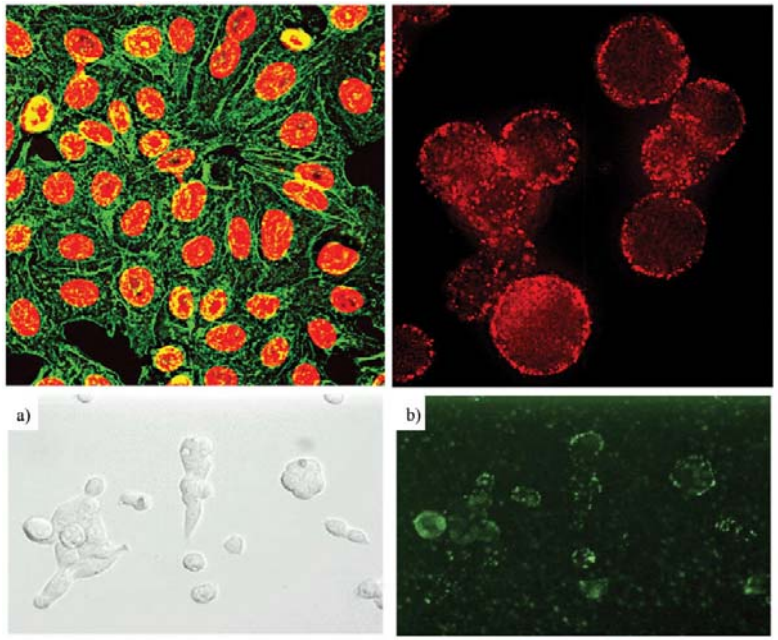
Solutions of monodisperse CdSe nanocrystals with increasing diameter (2 to 7 nm) from left to right





4 nm = 0.000004 mm

Colour changes with Size!

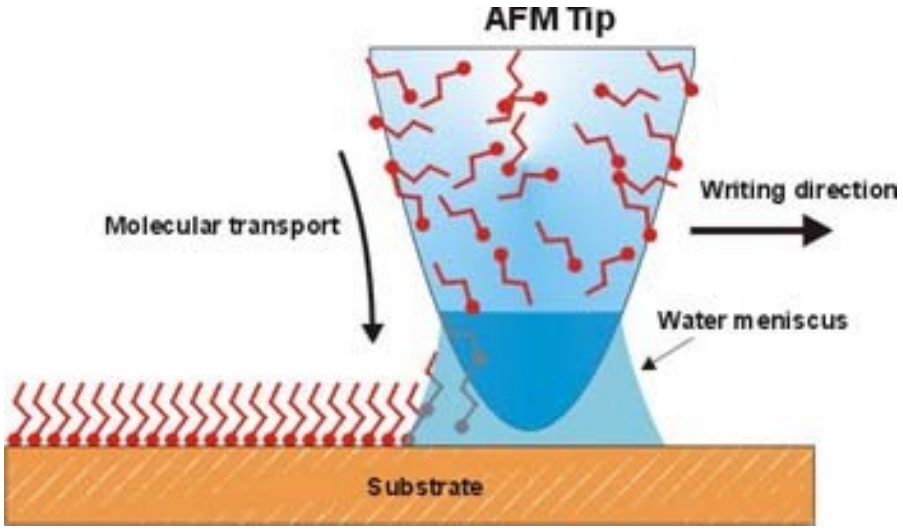


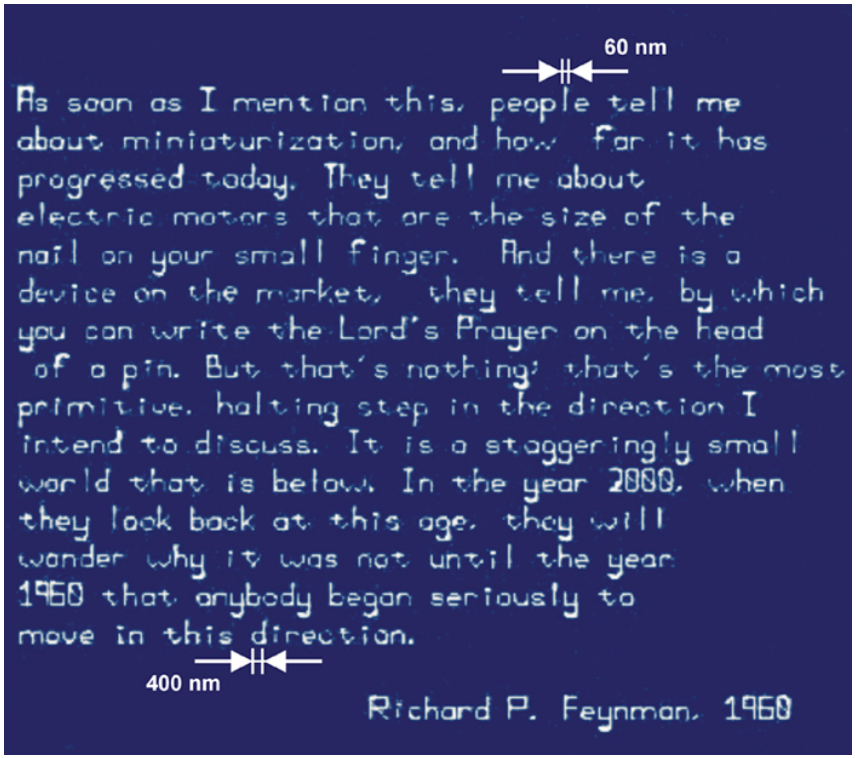
Parak et al., Nanotechnology **16**, R9 (2005)

Dip-Pen Nanolithography (DPN)

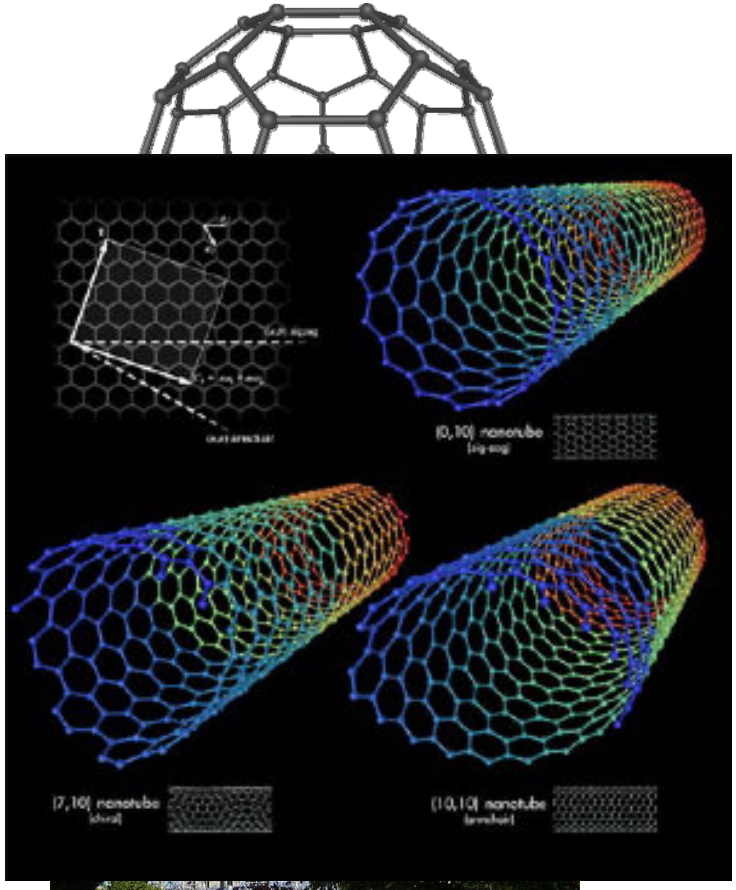


Chad Mirkin (Northwestern University)





Buckminsterfullerene (C_{60}) and Carbon Nanotubes

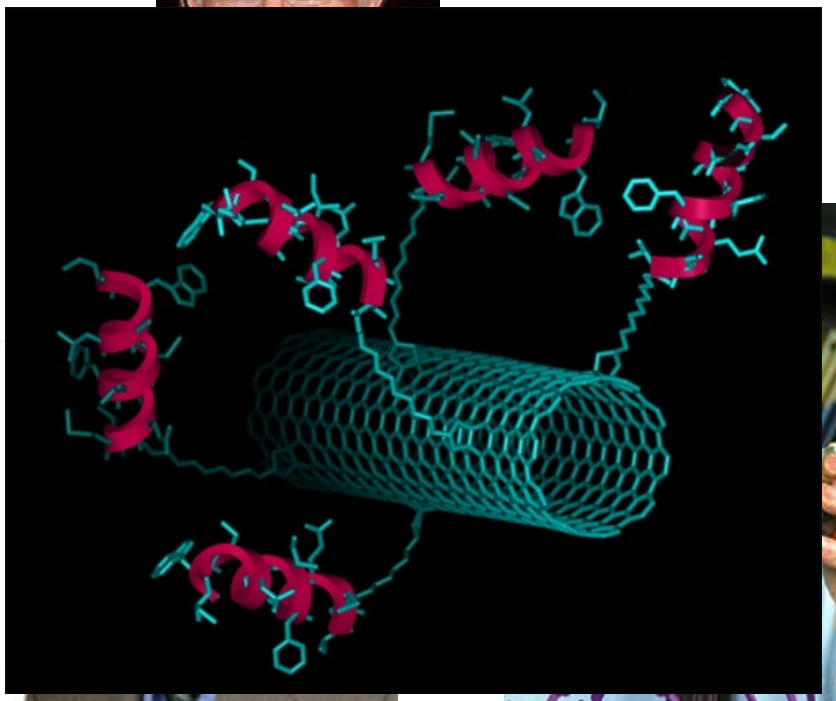


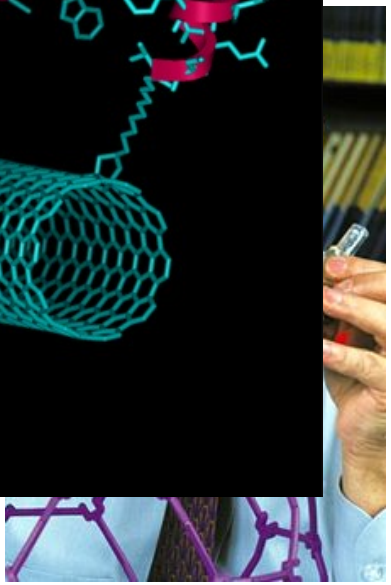


Geodesic dome by
R. Buckminster Fuller



Harold Kroto





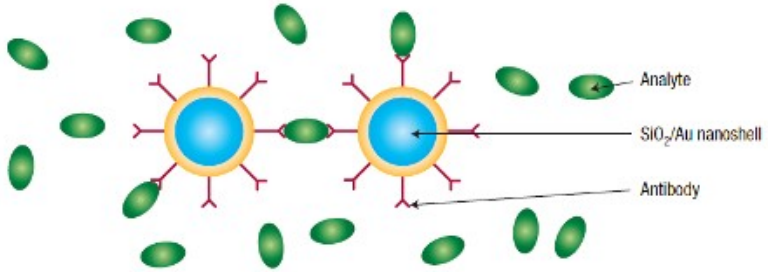
Robert Curl

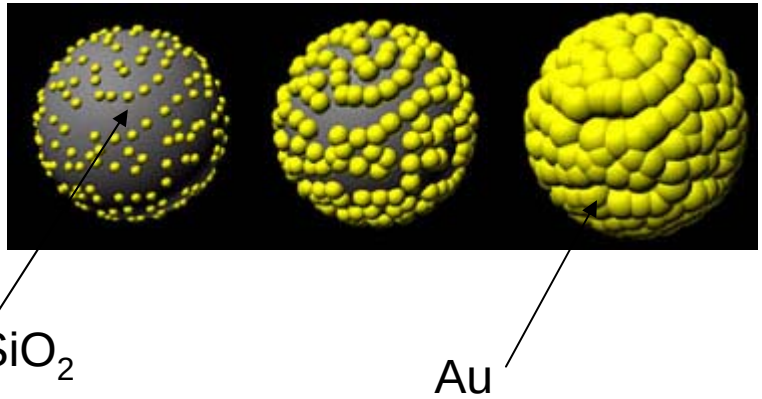
Richard Smalley

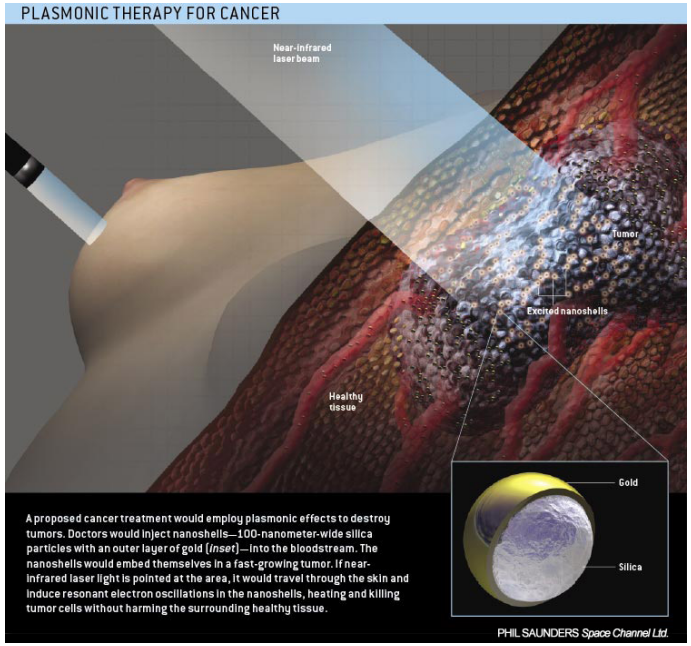
Nanoshells



Naomi Halas (Rice Univ.)



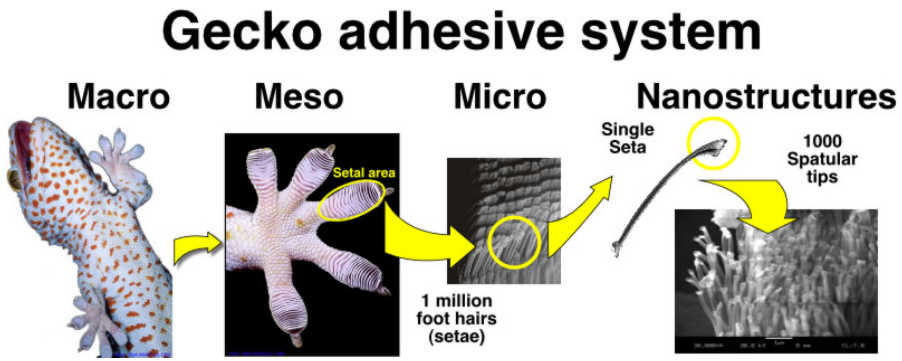




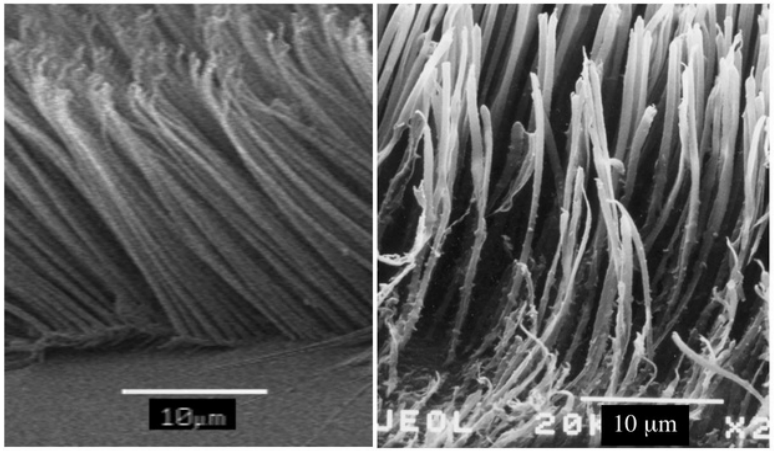
Nanoshells and breast cancer

Gecko and van der Waals Force

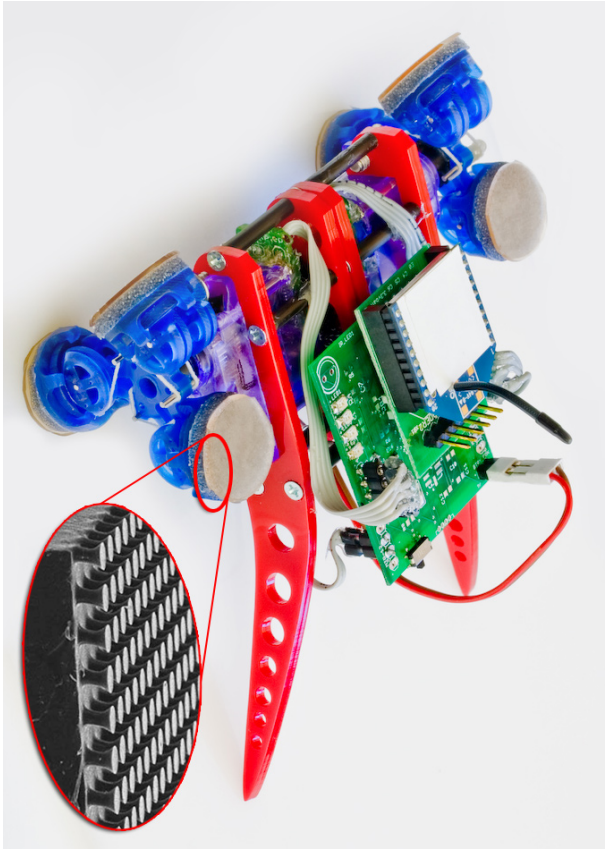






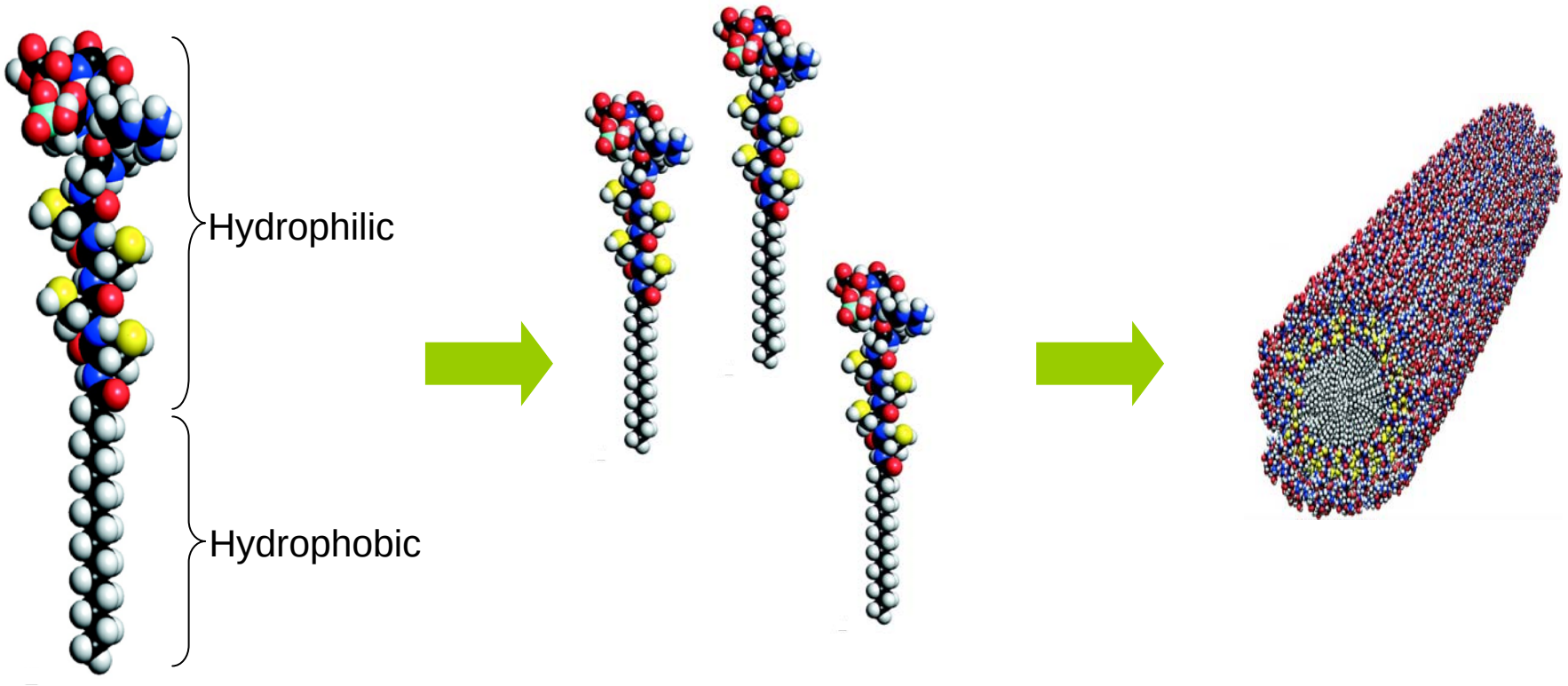


Gecko and van der Waals Force





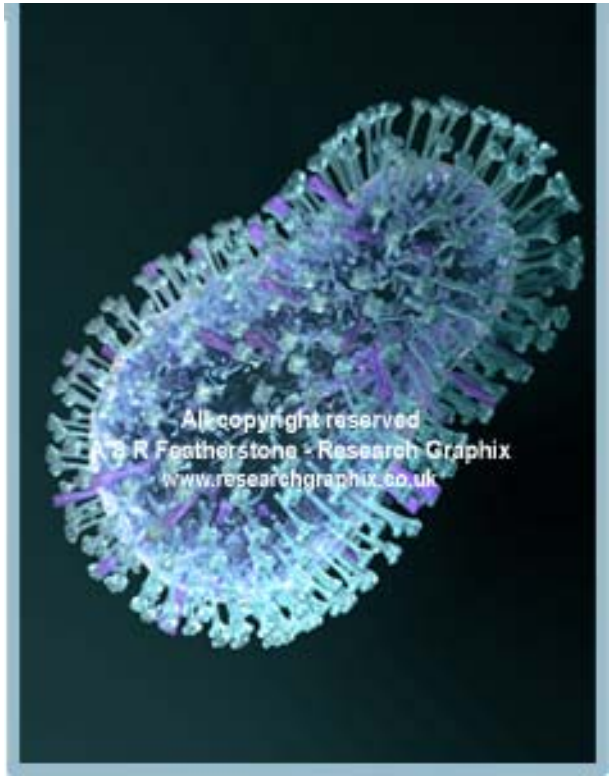
Self-assembly of peptide amphiphile nanofibers



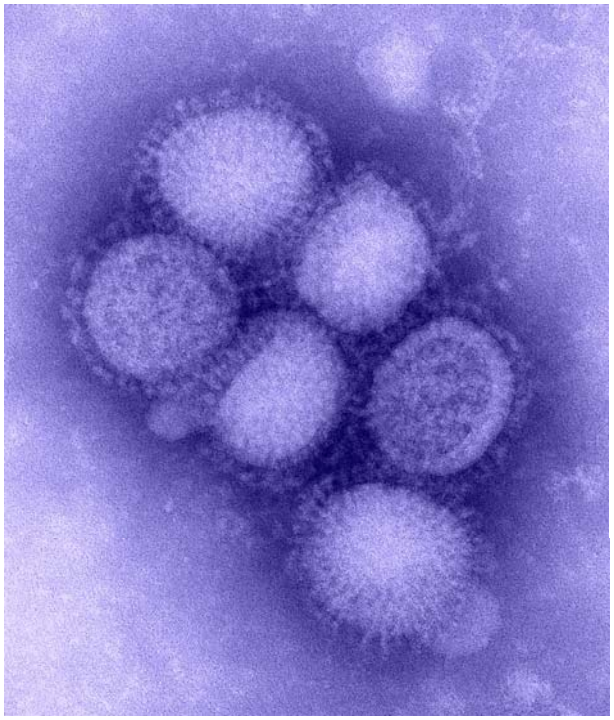
Hartgerink et al., Science **294**, 1684 (2001)

Avian Influenza (AI) and Swine Influenza (SI)

H5N1 variety of influenza A

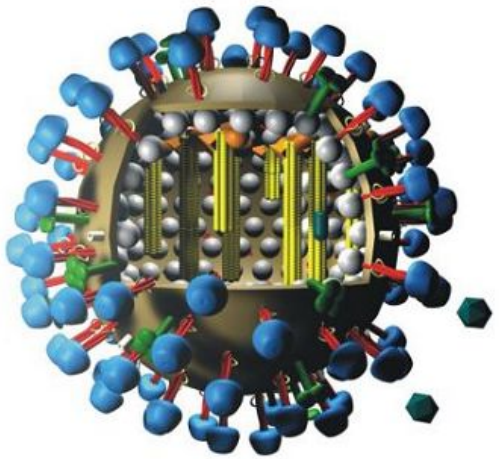


H1N1 variety of influenza A

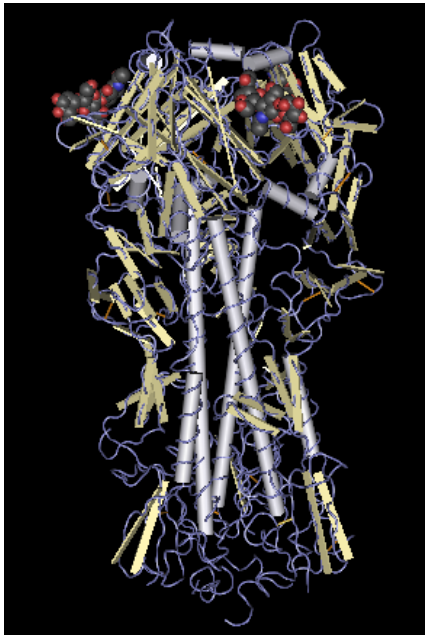


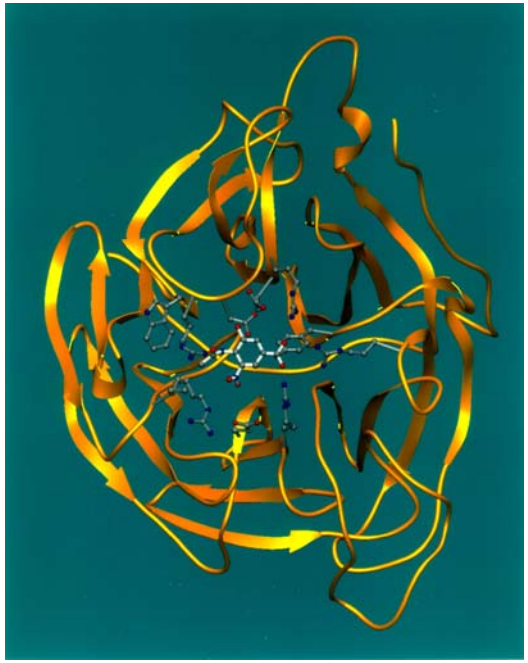
- Influenza A viruses are common in bird flu
- SI virus strains include influenza C and the subtypes of influenza A known as **H1N1**, H1N2, H3N1, H3N2, and H2N3.

Hemagglutinin and Neuraminidase



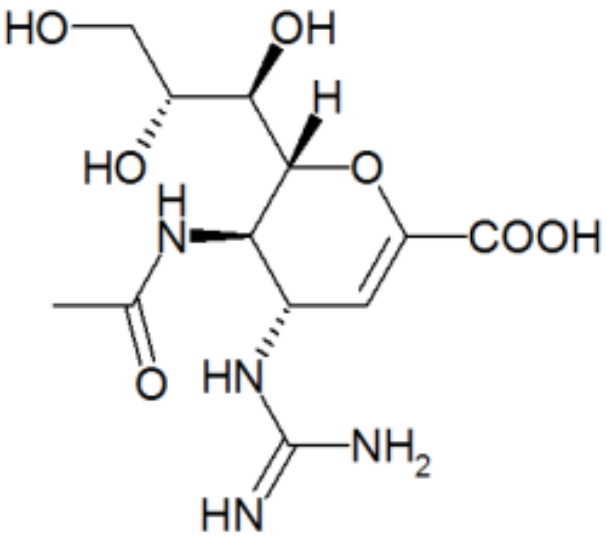
Hemagglutinin (HA)



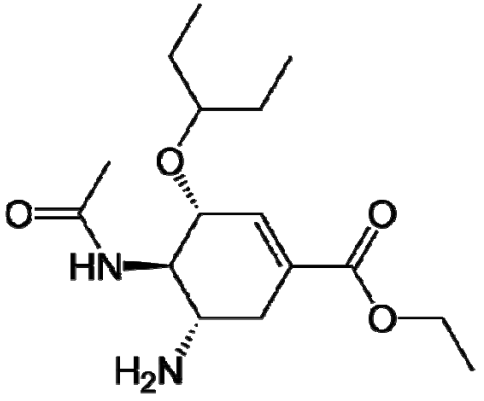


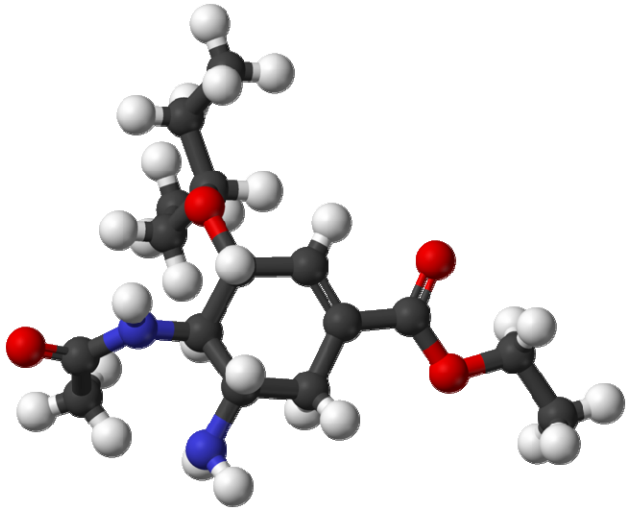
Neuraminidase (NA)

Zanamivir and Oseltamivir – antivirul drugs



Zanamivir (Relenza, GlaxoSmithKline)





Oseltamivir (Tamiflu, Roche)